

Integrated Robotic Actuator Technical Design Document

Electromechanical architecture, control design, verification, and integration plan

PROJECT

Integrated Robotic Actuator

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Abstract

This document captures the architecture, requirements, safety constraints, implementation notes, and verification plan for the NervTech Integrated Robotic Actuator project.

Keywords: robotic actuator, BLDC, embedded control, torque sensing, mechatronics

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REVISION HISTORY

Rev.	Date	Author	Change Description
A	2026-06-25	NervTech Engineering	Initial technical document class demonstration.

1 SYSTEM OVERVIEW

System Summary

Architecture	Compact BLDC actuator with gearbox, encoder feedback, and embedded motor controller.
Control	Cascaded current, velocity, and position control with torque-limit supervision.
Sensing	Motor phase current, rotor position, output position, temperature, and bus voltage.
Safety	Fused DC input, firmware watchdog, current limit, thermal derating, and emergency stop path.

Design Note – Architecture Intent

The integrated actuator should be documented as a full mechatronic system: motor, transmission, bearings, enclosure, PCB, firmware, control loops, and validation fixtures.

2 REQUIREMENTS

Table 1: Requirements

ID	Type	Requirement	Priority	Verification
IRA-REQ-001	Functional	The actuator shall support closed-loop position control using encoder feedback.	High	Bench test
IRA-REQ-002	Safety	The actuator shall enter a disabled state if bus voltage, phase current, or temperature exceeds configured limits.	Critical	Fault injection
IRA-REQ-003	Interface	The actuator shall expose telemetry over a deterministic digital interface.	Medium	Integration test

3 ELECTRICAL ARCHITECTURE

Safety Critical – Power Bring-up

Start with a current-limited bench supply, no load, conservative firmware limits, and an accessible emergency disconnect before energising the motor stage.

Table 2: System Interfaces

Interface	Domain	Signal / Data	Constraint
DC Bus	Power	24–48 V input, fused	Ripple and inrush must remain within controller limits
Encoder	Signal	Quadrature or SPI absolute position	Shielding and grounding required near motor phases
Host Link	Data	CAN, RS-485, or UART telemetry	Deterministic update period for control logging

4 FIRMWARE EXAMPLE

```
1 if (phase_current > current_limit || motor_temp > temp_limit) {  
2   controller.disable_pwm();  
3   fault_latch = true;  
4   publish_fault_status();  
5 }
```

Listing 1: Torque-limit supervision pseudocode

5 VERIFICATION PLAN

Table 3: Verification Test Cases

Test ID	Procedure	Acceptance Criteria	Result
IRA-TST-001	Command a 10 degree step response at no load.	Settling time and overshoot are within design target.	TBD
IRA-TST-002	Inject over-current threshold in firmware test mode.	PWM is disabled and fault is latched.	TBD